

Exercise Sheet 2

The Simplex Method and Diameter of Polyhedra

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This sheet restates the four exercises of Lecture 2. Throughout, $P = \{x \in \mathbb{R}^n : \mathbf{A}x \leq b\}$, with $\mathbf{A} \in \mathbb{R}^{m \times n}$ of full column rank and rows $a_1^\top, \dots, a_m^\top$, denotes a polyhedron in geometric form; a *vertex* of P is a feasible point with n linearly independent tight constraints. We will use the optimality certificate of Lecture 1: a vertex maximizes an objective over P if and only if the objective is a nonnegative combination of its tight constraint normals. A *connected layer family* (CLF) of order d on N symbols is a sequence $L_1, \dots, L_\ell$ of nonempty collections of d -element subsets of the ground set, all sets appearing in the family distinct, such that for all $i \leq j \leq k$ and all $A \in L_i, C \in L_k$ there exists $B \in L_j$ with $B \supseteq A \cap C$; its *length* is ℓ . As in the proof of the Kalai–Kleitman bound, $h(d, N)$ denotes the maximum length of a CLF of order d on N symbols, with $h(d, N) := 0$ if no such family exists; recall $h(1, N) \leq N$, the sets of an order-1 family being distinct singletons.

Exercise 1 (Removing degeneracy by perturbation). Assume that $P = \{x \in \mathbb{R}^n : \mathbf{A}x \leq b\}$ is pointed, i.e., $\text{rank}(\mathbf{A}) = n$, and that the LP $\max\{c^\top x : x \in P\}$ is feasible and bounded. For $\varepsilon > 0$, let $\varepsilon^{(m)} := (\varepsilon, \varepsilon^2, \dots, \varepsilon^m)^\top$ and consider the perturbed polyhedron $P_\varepsilon := \{x \in \mathbb{R}^n : \mathbf{A}x \leq b + \varepsilon^{(m)}\}$.

- (a) Show that there is $\varepsilon_0 > 0$ such that for all $0 < \varepsilon < \varepsilon_0$, every vertex of P_ε has exactly n tight constraints.
- (b) Prove that there exists $J \subseteq [m]$, indexing n linearly independent rows, such that $v_\varepsilon := \mathbf{A}_J^{-1}(b_J + \varepsilon_J^{(m)})$ is an optimal vertex of $\max\{c^\top x : x \in P_\varepsilon\}$ for arbitrarily small $\varepsilon > 0$. Show that, for any such J , the point $v_0 := \mathbf{A}_J^{-1}b_J$ is an optimal vertex of the original LP.

In particular, running the simplex method on the perturbed LP — where no cycling can occur — recovers an optimal tight-row set, and hence an optimal vertex, of the original LP. The perturbation can moreover be implemented *symbolically*: compare slacks, which are polynomials in ε , by the lexicographic order of their coefficients, i.e., by their behavior as $\varepsilon \rightarrow 0^+$ (the classical *lexicographic pivoting rule*).

Exercise 2 (Goldfarb cube). Let $0 < \beta < \alpha/4 < 1/8$, and consider $\max x_n$ subject to $0 \leq x_1 \leq 1$, $\alpha x_1 \leq x_2 \leq 1 - \alpha x_1$, and

$$\alpha(x_{k-1} - \beta x_{k-2}) \leq x_k \leq 1 - \alpha(x_{k-1} - \beta x_{k-2}), \quad 3 \leq k \leq n.$$

(Here it is convenient to run the shadow vertex rule in maximization form; equivalently, negate both objectives.)

- (a) Show that the feasible region is a combinatorial cube: at every vertex exactly one constraint of each pair is tight, and the 2^n vertices are indexed by the sets $S \subseteq [n]$ of pairs whose upper constraint is tight. (Hint: Prove by induction on k that every feasible point satisfies $x_k \geq \frac{\alpha}{2} x_{k-1} \geq 0$ and $x_k \leq 1$. Conclude that both constraints of a pair are never tight simultaneously, and that the tight systems are triangular.)

- (b) Show that every vertex is the unique maximizer of some objective $\gamma e_{n-1} + \delta e_n$. (Hint: By the certificate fact, it suffices to write $\gamma e_{n-1} + \delta e_n$ as a positive combination $\sum_k y_k a^{(k)}$ of the n tight constraint normals $a^{(k)} = \pm e_k + \alpha e_{k-1} - \alpha\beta e_{k-2}$ (with $e_0 = e_{-1} := \mathbf{0}$). Encode the weights by the ratios $s_k := y_k/y_{k+1}$: cancelling coordinate $k \leq n-2$ determines s_{k+1} from s_k . Choose $s_1 \in (0, \alpha/2]$ and propagate backwards, showing — using $\beta < \alpha/4$ — that every ratio required along the way stays in $(0, \alpha/2]$; finally unwind the weights from $y_n := 1$.)
- (c) Conclude that the shadow vertex rule interpolating towards e_n can be forced to visit all 2^n vertices. (Hint: By locality, a vertex is visited as soon as it uniquely maximizes some intermediate objective; choose the starting objective so that all the objectives (γ, δ) produced in part (b) occur along the interpolation.)

Exercise 3 (CLF calculus). Let $L_1, \dots, L_\ell$ be a CLF of order d on N symbols. Prove the two closure properties used in the lecture:

- (a) (*Taking sublayers.*) For all indices $i_1 < i_2 < \dots < i_r$, the subsequence $L_{i_1}, \dots, L_{i_r}$ is a CLF of order d .
- (b) (*Restricting to a common set.*) Let S be a set of at most $d-1$ symbols. The layers containing at least one set $\supseteq S$ form a contiguous (possibly empty) interval of $\{1, \dots, \ell\}$. Restricting these layers to their sets containing S , and deleting the symbols of S from each, yields a CLF of order $d - |S|$ on at most $N - |S|$ symbols, of the same length as the interval.

(Hint: For (b): if $A \supseteq S$ appears in layer i and $C \supseteq S$ in layer k , then for every $i \leq j \leq k$ connectivity provides $B \in L_j$ with $B \supseteq A \cap C \supseteq S$. Deleting the symbols of S keeps the sets distinct, since the sets involved all contain S .)

Exercise 4 (Proof of the Barnette–Larman bound). Let $L_1, \dots, L_\ell$ be a CLF of order $d \geq 2$ on N symbols. For each symbol s , let $J_s \subseteq \{1, \dots, \ell\}$ be the interval of layers containing a set through s , as in Exercise 3(b).

- (a) Greedily partition the layers into intervals: having covered $L_1, \dots, L_t$, choose the symbol s occurring in layer L_{t+1} whose interval J_s extends furthest to the right, and let the next interval be $\{t+1, \dots, \max J_s\}$. Conclude from Exercise 3(b) that each interval produced has length at most $h(d-1, N_k)$, where N_k is the number of symbols occurring in its layers.
- (b) Show that each symbol occurs in the layers of at most *two* consecutive intervals of the greedy partition. (Hint: Contiguity and the greedy choice.)
- (c) Conclude that $\sum_k N_k \leq 2N$, and by induction on d that $\ell \leq 2^{d-1}N$.